

Aylin Onalan

Professor Ong

CSS545

17 November 2024

Bridging the Gap in Cross-Platform Compatibility

Introduction

In today's world we have a diverse ecosystem of different devices and platform types, ranging from mobile phones, IoT devices, laptops, and more. Due to this, many applications need to be able to seamlessly function across multiple mediums, especially between mobile devices and laptops. Cross platform development tools are a resource that are able to bridge the gap between these different platforms, by creating user interfaces (UI) that provide quality user experience (UX).

Industry Trends and Needs

Cross-platform development is a relevant and critical concern in the software industry, with developers attempting to create applications that are capable of functioning consistently across different types of devices and operating systems. With the increase of smartphones, tablets, desktops, and wearable devices, the demand for applications that deliver seamless user experiences across platforms like Android, iOS, Windows, and macOS has increased significantly as well. The cross-platform development industry is currently experiencing an increase in Artificial Intelligence (AI) and Machine Learning (ML), with these technologies providing features such as natural language processing (NLP), custom tailored recommendations, and predictive analytics [1].

Current Solutions

One example of a prominent cross-platform development framework includes the Facebook created React Native, which helps developers construct mobile applications using the React and JavaScript programming languages [2]. React Native can provide almost identical experiences across different platforms like web and mobile, using a singular codebase for development platforms like iOS and Android, by translating the components from React into ones for native [2]. React Native has been used in order to create applications such as Airbnb, Instagram, and Facebook [2]. Another example of a cross-platform development framework includes Flutter, which was created by Google [2]. Flutter uses Dart as a programming language, and unlike the React Native platform, has an architecture based on widgets [2]. Notable examples of applications that have been developed using Flutter include Google Ads, Alibaba, and Reflectly [3]. Microsoft has its own platform development tool that it has acquired, by the name Xamarin, which uses the frameworks of C# and . NET [2]. Examples of applications that

have been developed using Xamarin include the Alaska Airlines application as well as Skyscanner [2].

Critical Analysis

Cross Platform Development Tool	Advantages	Disadvantages
React Native	React Native has a structure that is component-based, which allows app developers to make UI components that can be reused [4]. Additionally, React Native has third-party plugin compatibility in addition to allowing developers to see a preview of changes they have made to their code via hot reloading, which helps to speed up the development process and make debugging easier [4]. Overall, React Native is cost effective, provides a speedy development process, and has significant community support [4].	React Native has a learning curve for developers who have not used it before, in addition to having limitations in its performance [4]. Furthermore React Native applications tend to have a large file size in comparison to applications that are fully native, and also is not a good resource for situations that require complex operations or computation [4].
Flutter	Flutter is able to create a UI that is aesthetically pleasing and has a low latency [2]. Additionally, Flutter is an open-source framework and also has a development cycle that is efficient and speedy that allows for app developers to see modifications they make to their code in real time [3].	Flutter has limited third-party libraries, in comparison to more established cross-platform development frameworks like React Native [3]. Flutter additionally has a learning curve for developers who are not familiar with the programming language Dart [3].
Xamarin	One significant advantage of Xamarin, it that it is compatible with the Microsoft Ecosystem [2]. Additionally, Xamarin is a free to use cross-platform development tool for fledgling startups, smaller companies, and individual developers [5].	Xamarin is not as popular of a tool in comparison to Flutter and React Native, which means that there are less experienced users and less of a community [5].

Mitigating Limitations of Existing Solutions.

In order to address the limitations of existing cross-platform development tools, I suggest the use of a model-driven approach. By using a model-driven approach, developers are able to build off of or include new platforms without having to be limited to preset options [6]. Additionally, using a model driven approach allows for the effective distribution of functionality across various platform types, and is also able to lower complexity and cost by using a singular

development environment to manage tasks [6]. Overall, a model driven approach is a holistic way to deal with the limitations of existing cross-platform development tools.

Conclusion

In conclusion, the need for seamless cross-platform functionality is becoming increasingly critical in today's diverse device ecosystem. Current cross-platform development tools such as React Native, Flutter, and Xamarin are able to address many demands of the industry. However, these tools also have their own sets of limitations, including restricted libraries, performance issues, and learning curves. A model-driven approach is a potentially promising solution to these limitations, as it allows developers to integrate new platforms, distribute functionality effectively, and reduce complexity and costs by leveraging a unified development environment. This holistic strategy can enhance the efficiency and scalability of cross-platform development, meeting the evolving demands of the software industry.

References

- [1] Nintriva, “The Future of Cross-Platform App Development: Trends and Predictions,” Medium. Accessed: Nov. 17, 2024. [Online]. Available: <https://medium.com/@marketingnintriva/the-future-of-cross-platform-app-development-trends-and-predictions-7f491acd3c40>
- [2] O. D. Segun-Falade, Olajide Soji Osundare, Wagobera Edgar Kedi, Patrick Azuka Okeleke, Tochukwu Ignatius Ijomah, and Oluwatosin Yetunde Abdul-Azeez, “Developing crossplatform software applications to enhance compatibility across devices and systems,” *Comput. Sci. IT Res. J.*, vol. 5, no. 8, pp. 2040–2061, 2024, doi: 10.51594/csitj.v5i8.1491.
- [3] “Flutter: Advantages, Disadvantages and Future Scopes,” GeeksforGeeks. Accessed: Nov. 17, 2024. [Online]. Available: <https://www.geeksforgeeks.org/flutter-advantages-disadvantages-and-future-scopes/>
- [4] “React Native: Features, Limitations, Pros and Cons,” WebShark Web Services. Accessed: Nov. 17, 2024. [Online]. Available: <https://medium.com/webshark-web-services/react-native-features-limitations-pros-and-cons-0a692dc566da>
- [5] MobileSocket, “Advantages and Disadvantages of Xamarin App Development,” Medium. Accessed: Nov. 17, 2024. [Online]. Available: <https://medium.com/@socketmobileinc/advantages-and-disadvantages-of-xamarin-app-development-c30d8e716e24>
- [6] J. Z. Blanco and D. Lucrédio, “A holistic approach for cross-platform software development,” *J. Syst. Softw.*, vol. 179, p. 110985, Sep. 2021, doi: 10.1016/j.jss.2021.110985.